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Student ID	251925403
Student Name	Mohamed Mohamed Hussein
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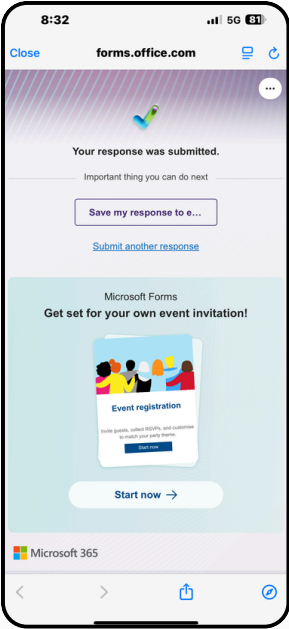
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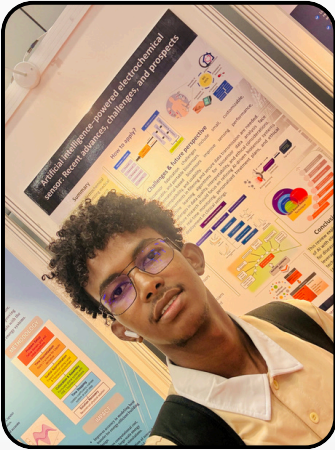
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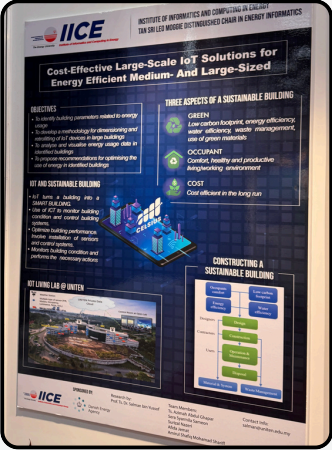
This is the screenshot confirming the registration for the Tan Sri Leo Moggie Lecture Series 2025 event.



This is the selfie taken inside the keynote speaker hall during the event.



This is the first of two selected IT-related research projects from the research exhibition.



This is the second selected IT-related research project from the exhibition.

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Keynote Lecture Summary: “Algae”

The keynote lecture by Professor Christopher Howe from the University of Cambridge was very interesting. The main topic was about algae, a very old and simple organism that can be used as a new solution for our energy problems. Professor Howe explained that we often have a bad image of algae because of things like green blooms in lakes, but it has a lot of positive potential.

My understanding from the talk is that algae can produce electricity through a process called photosynthesis. When algae are in the light, they create energy, and a small part of this energy can be collected as electricity. This process is called **biophotovoltaics (BPV)**. It works like a small, living solar panel. The lecture showed a simple diagram of a BPV cell with two parts, an anode and a cathode, where the photosynthetic material (algae) releases electrons to create an electrical current. Professor Howe showed that this electricity can be used to power small devices. He also talked about another use for algae, which is to produce biofuels. Some types of algae create oils called **triacylglycerides (TAG)**, which can be chemically changed into Fatty Acid Methyl Esters (FAME). FAME is a type of biodiesel that can be used in vehicles.

One of the main issues faced is that generating power from algae is not very strong. The amount of electricity is small, so it cannot be used to power big things like our homes or factories. It is only suitable for devices that use very little power, like small sensors or simple electronics. For biofuels, a big issue was that it was very expensive to produce. Professor Howe showed a slide about how major oil companies like ExxonMobil invested a lot of money in algae biofuel research but eventually stopped because it was not commercially successful yet. This shows that making it affordable is a big challenge.

Modern technology and IT play a big role in making algae a real solution. First, technology helps improve the BPV cells. The lecture showed examples of special electrodes, like 3D-printed micropillar arrays, that can collect the electricity from the algae more efficiently. Also, like the Internet of Things (IoT) is a big influence. IoT involves billions of small devices that need a small, continuous amount of power. Algae-powered cells are a perfect solution for this. Professor Howe gave an amazing example where his team powered an ARM microprocessor, a small computer chip, for over a year using only an algae cell. This is a great IT application because it replaces batteries that need to be changed and thrown away. The technology can also be very simple and low-cost. One slide showed how a BPV cell could be

made from recycled materials like a plastic bottle and shredded aluminum, making it accessible and sustainable.

In conclusion, the lecture taught me that even simple things in nature, like algae, can offer amazing solutions to big problems. While it is not powerful enough to replace traditional power plants, algae energy, with the help of modern IT and material science, has a very important future in powering the growing world of small, smart devices.

Speaker Observation Summary

Professor Christopher Howe appeared as a professional and experienced academic, dressed formally in a dark suit and tie which gave him a credible and authoritative presence. His presentation style was clear and effective, using well-designed slides with powerful images and diagrams, rather than too much text, to explain complex scientific topics visually.

I enjoyed the professor's physical delivery. He was very active and engaging while presenting content. He used his hands very naturally, and he showed how he felt about what he was saying; that dynamic also made the information interesting and engaging with the subject. He didn't just stand behind the podium, even though he is spaced out from the audience, he moved about in a seamlessly confident way stroll back and forth across the stage, which is a good technique to keep the audience's attention on him. His engagement with the audience seemed high as he took a complex scientific topic and presented it simply enough for his audience to follow along. Taking a very old species of species of algae, from billions of years ago, and connecting to very current problems like the internet of things, made the topic relevant and interesting to everyone in the auditorium.

Research Exhibition Summary

Research Project 1: Artificial Intelligence-Powered Electrochemical Sensor

The first study describes the power of Artificial Intelligence (AI) and electrochemical sensors, with a focus on healthcare and environmental monitoring. The core goal of this research review is to investigate the several ways that AI and current biosensors can combine to create groundbreaking new devices. The overview section describes that this combination is helping transform the development of pharmacological treatments, enhancing patient data collection,

and facilitating wearable sensors of personal health & fitness. One of the primary goals of the study is to introduce how these state-of-the-art sensors can detect specific “biomarkers” that suggest the presence of a particular illness so the health worker can provide more personalized and accurate “health care.”

The core technology is the integration of AI with electrochemical biosensors. The poster explains that the sensor detects a chemical or physical signal, and then AI algorithms—specifically mentioning Artificial Neural Networks (ANNs)—process the complex data to identify patterns, like how the human brain works. The research also highlights future challenges, such as the need for secure data transmission and addressing the ethical issues of using such devices in clinical settings. The research is conducted by **Dr. Siti Nur Ashakirin Binti Mohd Nashruddin** and her team.

My takeaway is that AI is becoming an essential tool for the future of medicine. The concept of a small, smart, wearable sensor that provides continuous, real-time health data is truly transformative. This could mean detecting serious illnesses like cancer much earlier than is possible today. However, the poster also made me think deeply about the challenges. As these devices collect very sensitive personal data, ensuring data privacy is extremely important. This project shows that while IT offers incredible life-saving potential, we must also be responsible for developing it ethically.

Research Project 2: Cost-Effective Large-Scale IoT Solutions for Energy Efficient Medium- and Large-Sized Buildings

This second research project, presented by the Institute of Informatics and Computing in Energy (IICE), proposes a solution to make medium and large-sized buildings more energy-efficient using the Internet of Things (IoT). The primary aim is to reduce energy waste, which contributes to a more sustainable environment. The specific objectives listed are, first, to identify the key parameters in a building that affect energy usage. Second, the project aims to develop a clear methodology for installing and setting up IoT services in large buildings. Finally, it seeks to analyze and visualize the energy usage data collected and then provide recommendations to optimize how energy is used in these buildings. The research emphasizes that a truly sustainable building must balance three aspects: being Green (low carbon footprint, energy efficient), providing a comfortable and healthy environment for the Occupant, and being Cost-effective in the long run.

The core technology is the Internet of Things (IoT), which turns a regular building into a **SMART BUILDING**. This is achieved by using a network of sensors and control systems to constantly monitor building conditions and performance. For example, sensors can track room temperature, occupancy, and light levels, and then the system can automatically adjust the air conditioning and lighting to save energy when no one is around. The research team is led by **Prof. Ts. Dr. Salman bin Yussof**.

My key takeaway is that this project provides a very practical and impactful application of IT. It shows a clear path to reducing a significant source of energy consumption in our cities. By making buildings smarter, we are not just adding technology for convenience; we are creating systems that actively contribute to environmental sustainability. This research is important because it moves beyond theory and demonstrates a real-world solution that can be implemented to make our infrastructure greener and more cost-efficient.